

Literature-Implied Partial Subcase for arXiv:math/0410256

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Status

Status: literature-implied answer, partial subcase.

The source paper is J. M. F. Castillo and Y. Moreno, *Extensions by spaces of continuous functions*, arXiv:math/0410256. In Section 4, item 6, the authors ask whether

$$\text{Ext}(M, C[0, 1]) = 0$$

for every subspace M of ℓ_1 .

The supporting paper is N. J. Kalton, *On subspaces of c_0 and extension of operators into $C(K)$ -spaces*, arXiv:math/9911144.

Recorded Subcase

Let M be a closed subspace of ℓ_1 . If M has an unconditional finite-dimensional decomposition, then

$$\text{Ext}(M, C(K)) = 0$$

for every compact Hausdorff space K . In particular,

$$\text{Ext}(M, C[0, 1]) = 0.$$

By Castillo–Moreno’s main theorem, such an M also admits a fully w^* -continuous scalar metric projection $Z(M, \mathbb{R}) \rightarrow M'$.

Identification

Kalton proves that a separable Banach space with a UFDD and the very strong Schur property is isomorphic to the dual of a subspace of c_0 . This is Theorem 5.3 in the arXiv PDF. A closed subspace of ℓ_1 inherits the strong Schur property from ℓ_1 . Kalton also records that, for closed subspaces of L_1 , the strong Schur property is equivalent to the very strong Schur property. Therefore a UFDD subspace $M \subset \ell_1$ is isomorphic to the dual of a subspace of c_0 .

Johnson–Zippin’s extension theorem, recalled in both the source paper and Kalton’s paper, says that spaces isomorphic to duals of subspaces of c_0 satisfy $\text{Ext}(X, C(K)) = 0$ for every compact K . Applying this theorem to the M above gives the stated subcase.

Scope and Remaining Problem

This is not a full solution of Castillo–Moreno’s question. It covers the UFDD branch of the subspace-of- ℓ_1 problem. The arbitrary closed subspace case remains open from the searches performed here, as does the broad question about metric projections continuous at zero for all spaces with $\text{Ext}(X, c_0) = 0$.

References

- J. M. F. Castillo and Y. Moreno, *Extensions by spaces of continuous functions*, arXiv:math/0410256.
- N. J. Kalton, *On subspaces of c_0 and extension of operators into $C(K)$ -spaces*, arXiv:math/9911144.
- W. B. Johnson and M. Zippin, *Extension of operators from weak*-closed subspaces of l_1 into $C(K)$ spaces*, Studia Math. 117 (1995), 43–55.